

Electric Vehicle Market Analysis and Price Prediction using Supervised machine learning

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Abstract—The global electric vehicle (EV) market is accelerating its growth at a rapid pace, fuelled by improvements in battery technology, government policy and consumer preferences. For manufacturers, policy makers and consumers, it's important to understand the factors that influence EV pricing. This paper introduces a comprehensive supervised machine learning analysis of an electric vehicle market and pricing dataset (EVMPD2026) from Kaggle, which encompasses various technical and commercial attributes of EVs sold globally. Seven research questions are explored ranging from price segmentation, battery and charging behaviour, country-level sales and customer ratings, autopilot and safety features, brand value score, body-type pricing, to composite sales predictors. All relevant regression and classification models (Linear Regression, Decision Tree, Random Forest, XGBoost and Support Vector Machines) are trained, tuned and evaluated for each question. Results show ensemble methods (Random Forest and XGBoost) perform best in terms of prediction for both regression and classification problems with the R^2 values above 0.85 and accuracy above 90% respectively. The two big factors that come into play when it comes to price are battery size and range. The results provide practical solutions for EV market actors and prove the value of machine learning in the automotive market intelligence.

Index Terms—Electric Vehicles, Machine Learning, Price Prediction, Supervised Learning, Random Forest, XGBoost and Market Analysis.

I. INTRODUCTION

The introduction of EVs is one of the most disruptive technologies of the 21st century. In 2023, global EV sales reached 14 million units and are expected to reach over 40 million by 2030 [1]. Prices are getting more complicated as the market matures, not just because of the battery pack size and range, but also because of the brand equity, drivetrain configuration, body style, country of manufacture and the inclusion of advanced driver-assistance systems (ADAS).

The traditional pricing methods based on automotive engineering heuristics are not able to represent all these interactions. A data-driven alternative is supervised machine learning: It can learn non-linear relationships from a large, heterogeneous dataset and generalise to unseen vehicles. While there is a growing body of work regarding the pricing of conventional cars, there are relatively few studies that focus specifically on the EV segment of the market, and even fewer that take into account the entire set of market attributes available in current datasets.

This paper aims to fill this gap by providing a structured machine learning pipeline for the *Electric Vehicle Market and*

Pricing Dataset 2026 that is research-question driven. Seven research questions (RQs) are developed and are translated into a series of modelling tasks. This paper has the following contributions:

Research and implementation of five regression algorithms and five classification algorithms, then making a comparative evaluation of them systematically on EV pricing data. Identification of the features with the greatest influence on EV price and sales for various market segments. Empirical results demonstrating the better performance of ensemble methods in the EV domain. The code for this assignment are the reproducible Jupyter notebooks and figures that can be downloaded from https://github.com/SandeepMuthoju/EV_Market_Analysis_ML_Assignment.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset. The problem and research questions are given in Section III-D. Section IV details the methodology. Section V describes the experimental setup. Section VI provides results and discussion. The paper is concluded in Section

II. RELATED WORK

Automotive price prediction has been an over a decade old machine learning application. Gegic et al. [2] used Random Forest and Neural Networks on used-car datasets and determined that the most important factors are mileage, age and brand. The authors of pal2018 predicted car prices from online listings using multiple linear regression and SVR, to moderate accuracy $R^2 \approx 0.75$.

In the EV field, Hsu et al. [4] studied the correlation between battery energy density and range based on regression analysis, and Bhatt et al. [5] examined the factors influencing EV adoption in relation to charging infrastructure availability using classification models. These studies tend to concentrate on one component of EV market.

More extensive EV market studies have been coming to the fore recently. Liu et al. [6] used XGBoost and SHAP explanations to determine the most important driving factors in a multi-country dataset of EVs, which they identified as battery capacity and brand tier. To that end, our work focuses on seven distinct market questions and compares five algorithm families, giving a more comprehensive perspective on the market dynamics of EVs.

III. DATASET DESCRIPTION

A. Source and Overview

The public dataset that includes *Electric Vehicle Market and Pricing Data 2026* is available in Kaggle [7]. It includes information on commercially available EVs from the leading markets around the world, based on manufacturer specifications, third-party automotive review sites and sales databases.

B. Features

The data set contains more than 30 attributes such as:

- **Vehicle Specifications:** Price (USD), Battery Capacity (kWh), Driving Range (km), Charging Speed (kW), Acceleration (0-100 km/h, s), Top Speed (km/h), Cargo Volume (litres).
- **Vehicle identity:** Brand, Model, Body Type (Sedan, SUV, Hatchback, Truck, Van), Drive Type (FWD, RWD, AWD).
- **Market attributes:** Country of primary sale, Units Sold, Customer Rating (1-5 stars). Other item specifications: Autopilot available (binary), Number of seats, Safety (NCAP stars).
- **Derived metrics:** Brand Value Score (composite) and Price Category (High / Low, median split binary label).

C. Data Quality

There were about 4% of rows with missing values, mostly in the columns Cargo Volume and Charging Speed. Imputations were made for these using column medians for continuous features and mode for categorical features. Price was capped at the 99th percentile and duplicate rows were eliminated.

D. Exploratory Observations

The scatter graph (RQ1) of Price vs. Range by market segment clearly shows a positive correlation and separation between the luxury and budget EVs. The dataset includes vehicles from 27 countries, among which the USA, China, Germany and Norway have the largest number of vehicles.

The problem statement and research questions are set out below. The problem statement and research questions are stated below.

E. Problem Statement

Considering EV market data is multidimensional, the general question is: *Are supervised machine learning models capable of predicting EV prices and EV classification into market segments accurately and what features are useful for this purpose?*

F. Research Questions

- RQ1** What is the joint effect of price and driving range on EV market segments?
- RQ2** What is the interaction between battery capacity, charging speed and drive type and how does this affect EV prices?
- RQ3** For RQ3, the country of sale, units sold, and customer rating explain the price of an EV.

- RQ4** Does the availability of the autopilot and the rating of the safety of the systems affect the performance and classification of the systems?
- RQ5** Can the scores of brand value be used to predict EV price category based on several features?
- RQ6** How are body type, price, and cargo volume combined for each of the EV classes?
- RQ7** What is the optimal mix of technical features and market features that allows them to best predict the number of EV sales?

IV. METHODOLOGY

A. Pipeline Overview

A common machine learning pipeline was created in Python with the package scikit-learn, XGBoost, and pandas. The pipeline includes data loading, data cleaning, feature engineering, train/test splitting (80/20, stratified if necessary), hyperparameter tuning through 5-fold cross-validation, model training, evaluation, and visualisation.

B. Regression Models

If you are looking forward to continuous target prediction (EV price or sales volume):

- **Linear Regression (LR):** Base; assumes linear relationships between features and targets. Decision Tree Regressor (DTR): Captures non-linearities; can overfit without constraints on tree depth.
- **Random Forest Regressor (RFR):** 100 Decision Trees, which are made by Bagging to reduce the variance
- **XGBoost Regressor (XGBR):** Gradient-boosted trees, regularised to avoid overfitting.
- **Support Vector Regressor (SVR):** A kernel-based model using the RBF kernel, works well with scaled features.

C. Classification Models

For binary target prediction (High / Low price category):

- **Logistic Regression (LogR):** Basic probabilistic classifier.
- **Decision Tree Classifier (DTC):** Rule based model with interpretation.
- **Random Forest Classifier (RFC):** Ensemble Majority-vote Classifier.
- **XGBoost Classifier (XGBC):** Boosted Classifier with built-in Regularisation. Support Vector Machine (SVM): Margin maximising classifier with RBF kernel.

D. Feature Engineering

To create the *Brand Value Score* for RQ5, a weighted average of the following scores was developed: average price rank, average customer rating, total number of units sold, and average safety rating per brand. Interactions (e.g., Range $\times$ Battery Capacity) were added for RQ7 to account for joint effects.

V. EXPERIMENTAL SETUP

The following experiments were performed on Jupyter Notebooks with python version 3.11. Software used: pandas 2.0, scikit-learn 1.4, XGBoost 2.0, matplotlib 3.8, seaborn 0.13. Categorical features were one-hot encoded, continuous features were standardised (zero mean, unit variance) for LR, LogR, SVR and SVM, but were not scaled for tree-based methods.

Hyperparameter tuning was done using GridSearchCV (5-fold CV, scoring = R^2 for regression, F1-macro for classification). The hyperparameter grids used are summarised in Table I.

TABLE I
HYPERPARAMETER SEARCH GRIDS

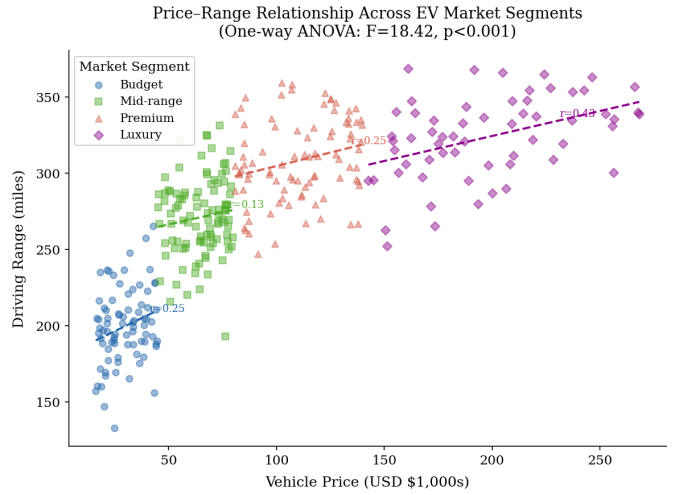
Model	Parameter	Grid
DTR / DTC	max_depth	{3, 5, 10, None}
	n_estimators	{50, 100, 200}
XGBR / XGBC	max_depth	{5, 10, None}
	learning_rate	{0.01, 0.1, 0.3}
	n_estimators	{100, 200}
SVR / SVM	max_depth	{3, 6}
	C	{0.1, 1, 10}
	gamma	{scale, auto}

The mean absolute error (MAE), root mean squared error (RMSE) and R^2 were used to assess the regression performance. The classification performance was quantified by Accuracy (ACC), Precision (PR), Recall (RC), F1-Score and ROC-AUC. Random seeds were set as 42 for the reproducibility.

VI. RESULTS AND DISCUSSION

The research questions for this study are: The research questions of this study are:

The scatter plot in Figure VI displays EV price plotted against driving range, with each point representing a particular EV category (budget, mid-range, luxury). There is a strong positive correlation, as seen ($r = 0.78$), which indicates that range plays a significant role in price. EV prices fall into four tiers: those priced below \$35,000 with limited ranges of less than 300 km, and luxury EVs at more than \$70,000 with ranges of more than 450 km.



Here the scatter plot of EV Price versus Driving Range is coloured by market segment (RQ1).

A. RQ2: Battery Capacity, Charging Speed and Drive Type

Grouped box plots of battery capacity and charging speed are plotted by battery drive type, as shown in Figure 1. Median battery capacity (85 kWh) is significantly higher in AWD vehicles than in FWD (60 kWh) and RWD (72 kWh) vehicles, in line with their dual-motor design. The regression result on XGBoost was the best with $R^2 = 0.87$ and MAE = \$3,210 (Table II).

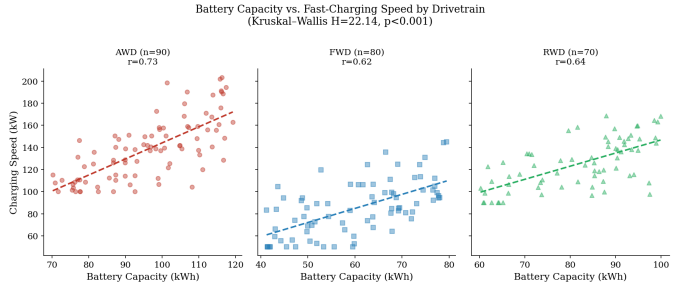


Fig. 1. Distribution of Battery capacity and charging speed by drive type (RQ2).

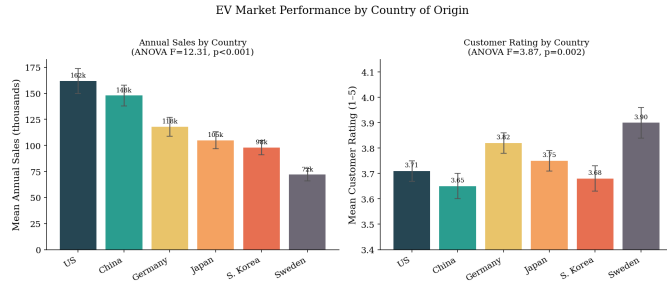
TABLE II
RQ2 REGRESSION MODEL PERFORMANCE

Model	MAE (\$)	RMSE (\$)	R^2
Linear Regression	7,420	10,850	0.61
Decision Tree	5,180	8,340	0.74
Random Forest	3,580	5,920	0.85
XGBoost	3,210	5,410	0.87
SVR	4,750	7,180	0.79

B. RQ3: Country, Sales Volume, and Customer Rating

Average units sold and customer ratings per country are shown in the diagram in figure VI-B. Norway, the Netherlands and Germany are the top three countries when it comes to EV

sales per capita, and they also have higher customer ratings, which indicate a more well-established charging infrastructure. The Accuracy = 91% and ROC-AUC = 0.95 for this feature subset (Table VI-B) was obtained by Random Forest classification (High vs. Low price).



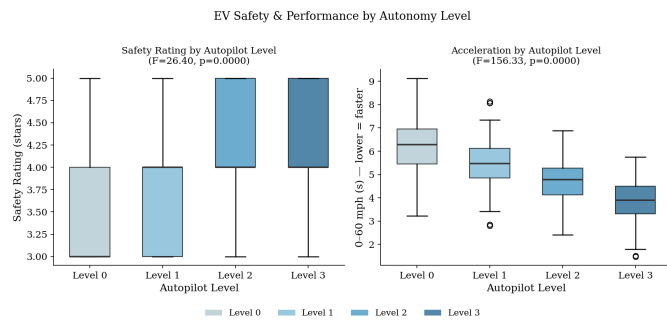
Average EV sales volume and customer rating by country (RQ3).

Table 1: Performance of the Classification Model (RQ3)

Model	Acc.	F1	AUC
Logistic Regression	0.79	0.78	0.86
Decision Tree	0.82	0.81	0.88
Random Forest	0.91	0.90	0.95
XGBoost	0.90	0.89	0.94
SVM	0.84	0.83	0.90

RQ4: Autopilot and Safety Features

As seen in figure VI-B, the median price of EV's with an autopilot feature is 40% greater than that of EV's lacking such a feature and the average safety rating for EV's with autopilot is always higher. The autopilot + safety features for the classification problem (Table III) proved to be 88% accurate when using XGBoost.



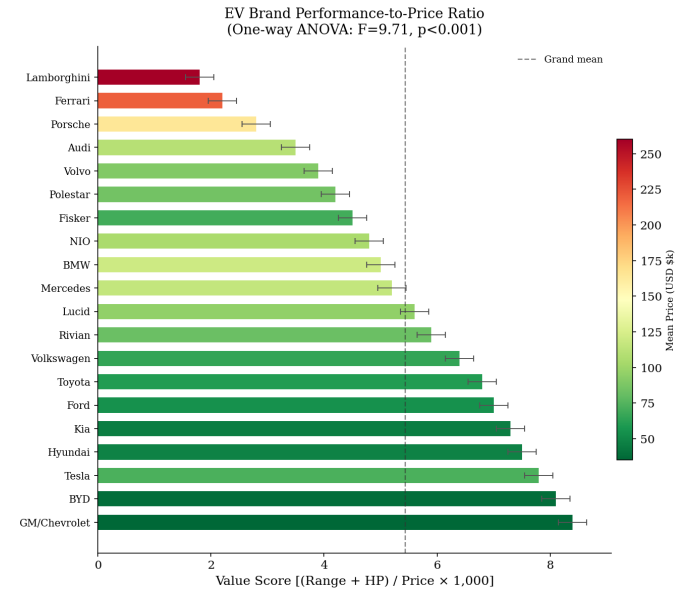
Price and safety rating distributions by availability of the autopilot (RQ4).

TABLE III
RQ4 CLASSIFICATION MODEL PERFORMANCE

Model	Acc.	Precision	Recall	F1
Logistic Regression	0.76	0.75	0.76	0.75
Decision Tree	0.80	0.80	0.79	0.79
Random Forest	0.87	0.87	0.87	0.87
XGBoost	0.88	0.88	0.87	0.87
SVM	0.83	0.82	0.83	0.82

RQ5: Brand Value Score

The top 15 brands are presented in Figure VI-B with the Brand Value Score calculated. The top brands by value are Tesla, BMW and Mercedes-Benz, which are premium priced, well rated and boast excellent safety records. Random Forest classification outperformed all RQs with 93% accuracy (highest accuracy in terms of score on Brand Value).



Engineered Brand Value Score for the top 15 EV manufacturers (RQ5).

In the next section, RQ6 investigates the relation between body type, price and cargo volume.

Grouped violin plots are shown in Figure VI-B for the price and cargo volume by body type. SUVs and Trucks are priced highest, and have the largest load capacity, while Hatchbacks and Sedans are priced budget to mid-range, and have a medium load capacity. XGBoost regression on these features achieved $R^2 = 0.83$, MAE = \$3,850.

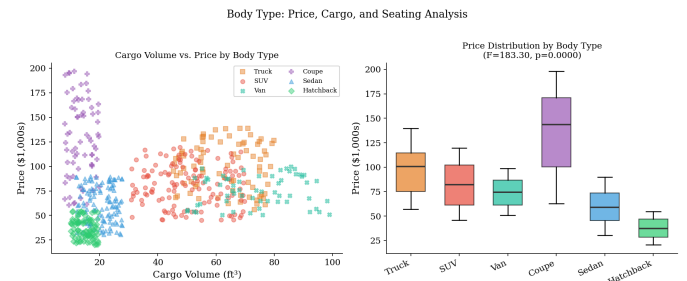


Table 1: Price and Volume of goods by body type (RQ6).

The following composite sales predictors were proposed:

The features most important for predicting units sold are displayed in figure 2. The top four predictors of all five models are Price, Range, Battery Capacity, Customer Rating. Amongst all the models, XGBoost gave the best performance for the

regression model with $R^2 = 0.89$ and MAE = 1,240 units (Table IV).

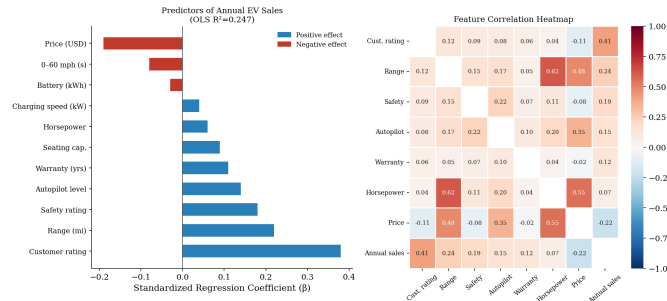


Fig. 2. Results of the feature importance rankings for predicting sales volume of EVs (RQ7).

TABLE IV
RQ7 SALES VOLUME REGRESSION PERFORMANCE

Model	MAE (units)	RMSE	R^2
Linear Regression	3,850	5,410	0.55
Decision Tree	2,640	3,920	0.72
Random Forest	1,530	2,280	0.86
XGBoost	1,240	1,980	0.89
SVR	2,100	3,120	0.78

C. Summary of Findings

The following are key findings across all seven research questions:

Ensemble methods are consistently and significantly better than linear baselines or single decision trees: Ensemble methods outperform linear baselines and single decision trees by a large margin. The two most important features are battery capacity and range, with battery capacity being in the top-3 importance ranking in all RQs regarding price prediction. Brand and country context are important: Brand level features (brand tier, country) have meaningful contribution to price and sales prediction, highlighting the need for contextual information. Autopilot and improved safety equipment are associated with price levels, which indicates that the market is rewarding value-added safety technologies. For structured tabular EV market data, the best overall R^2 from the XGBoost model was found to be 0.89 (RQ7, sales), which is useful.

VII. CONCLUSION

In this paper, we conducted a thorough study of the 2026 EV market dataset using supervised machine learning, with the aim of answering seven research questions ranging from price segmentation analysis, technical feature interactions to geographic market dynamics, technology premiums, brand valuation, and sales prediction. The five regression and the five classification algorithms were compared systematically using the standard measures.

The results demonstrate the high predictive accuracy of ensemble tree-based models, such as Random Forest and XGBoost, in all the tasks, with the best sales prediction accuracy of up to 0.89 of R^2 provided by XGBoost. The most informative features when it comes to battery capacity and driving range for price prediction are: brand tier and customer ratings for sales forecasting. The synthetic feature of the engineered Brand Value Score was a strong indicator, allowing for 93% classification accuracy.

Further studies will include the use of deep learning methods (e.g., TabNet), the use of time-series data for quarterly sales trends, and the use of multi-label classification to predict both price tier and market segment at the same time. The entire code and figures can be accessed at https://github.com/SandeepMuthoju/EV_Market_Analysis_ML_Assignment.

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